

Seeding QDArchive

Part 2: Data Classification

Project Report — Applied Software Engineering Project

Repositories	2
Projects classified	352 of 355 in scope
Primary data files classified	9,939 of 9,946
Taxonomy	ISIC Rev. 5 — section and division
Dominant class	85 Education (189 projects)

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1. Introduction

QDArchive is a web service for researchers to publish and archive qualitative data, with an emphasis on qualitative data analysis (QDA) files. Because the service is new, it has to be seeded with openly licensed qualitative data that already exists on the web. Part 1 of this project acquired that data; this report covers Part 2, which classifies it.

Part 2 has three requirements. Every project must be assigned a `PROJECT_TYPE` derived from the types of the files it contains. A classifier must then be developed that uses both the base data and the metadata, and that classifies against the ISIC Rev. 5 standard two levels down, i.e. to division level. Finally the classifier must be run by project type and by repository, and the resulting distributions reported.

1.1 Data provenance

The input is the Part 1 database `23025328-seeding.db`, tagged `part-1-release`. It holds 387 projects and 11,979 file records from 2 assigned repositories: the Harvard Murray Research Archive and the Australian Data Archive (ADA). Part 1 is treated as read-only input; all Part 2 output is written to a separate database, so the acquisition results remain reproducible and unmodified.

Of the 11,979 file records, 2,258 files were actually downloaded; the remainder are restricted and were recorded by Part 1 with a failure status rather than silently dropped. This asymmetry shapes the classifier design in Section 2.

2. Method

2.1 Merging and deduplication

The seeding database is merged into a new classification database. Deduplication treats the DOI as the global identity of a project, falling back to the project URL and then to the repository-scoped title. Merging further student databases requires only listing them in the configuration; the results reported here use this student's data, in which no duplicates were found.

2.2 Deriving the PROJECT_TYPE

Each project is typed from the types of its files, following the rules of the task description in strict precedence:

PROJECT_TYPE	Rule
QDA_PROJECT	There is a file with a QDA file extension
QD_PROJECT	Not a QDA_PROJECT and there are primary data files
OTHER_PROJECT	Not a QD_PROJECT and there are valid data files
NOT_A_PROJECT	Nothing can be derived about the file types

Table 1: The PROJECT_TYPE derivation rules.

The file taxonomy is keyed on the file extension. About 200 files in the Murray archive have no usable extension, either because none was ever given (RM 2462 ELias) or because a stray character was typed into it (...PDF´, .Paris-f-38_Draftsmanpdf). Since Part 1 also recorded the MIME subtype reported by the repository, the category falls back to it when the extension is unusable. The task description asks for the type to be derived from the file types, and that column is a file type, so this is the intended signal rather than a workaround. It recovers every one of those files — 73 of them PDFs, i.e. primary data — and corrects two projects that would otherwise have been misreported as NOT_A_PROJECT.

2.3 Classification taxonomy

The taxonomy is ISIC Rev. 5, taken from the structure adopted by the United Nations Statistical Commission at its 54th session. It is hierarchical: 22 sections identified by a letter, and 87 divisions identified by two digits. As required, classification goes down two levels, so every classified project carries a division and, through it, a section. The full taxonomy is embedded in the delivered database in the ISIC_CLASSES table, so the database is self-describing.

2.4 Classifier design

The classifier is a rule-based ISIC lexicon scored over a TF-IDF weighting. It is deterministic and involves no external service, so the results can be reproduced exactly by re-running the code. It works in four stages:

- Every project becomes one document built from its metadata (title, keywords, description), its file names, and the text extracted from the primary data files that were downloaded. A project is classified as the sum of its files, so the file text is concatenated.
- The documents are vectorised with a TF-IDF model restricted to the lexicon vocabulary of 729 terms spanning all 87 divisions. TF-IDF earns its place because the corpus is boilerplate-heavy: the phrase “the Murray Archive holds...” occurs in roughly 85% of the descriptions, and inverse document frequency drives such terms towards zero on its own, without a hand-maintained stop list.
- Each division scores as the sum of the TF-IDF weights of its matched lexicon terms, multiplied by a hand-assigned term weight that separates unambiguous evidence (“nursing home”) from weak evidence (“training”).

- The highest-scoring division becomes the primary class. The runner-up becomes the secondary class only when it reaches 45% of the top score and belongs to a different ISIC section, so that a class is not shadowed by a near-duplicate of itself.

The metadata score and the file-text score are computed separately and then blended, weighted 65% to 35% in favour of the metadata. Concatenating the two into a single document would let a 20,000-character codebook contribute hundreds of terms and bury a five-word title. Because only one file in six could be downloaded, a project with no file text would otherwise score systematically lower than one with text; such projects are therefore scored on their metadata alone rather than against zero, which keeps them comparable.

A project whose best division does not clear a minimum score is left unclassified rather than forced into a division, following the instruction to leave a field empty when it cannot be filled. Search tags are generated per project by a second, free-vocabulary TF-IDF pass and stored in the delivered database.

2.5 What is being classified

ISIC classifies economic activities, whereas the objects being classified here are research projects. A project is therefore assigned the division of the activity its data is *about*, not the activity of doing research. The alternative reading would collapse the entire archive into division 72 (Scientific research and development) and the taxonomy would carry no information at all; division 72 is accordingly reserved for projects whose subject really is research and development itself. This decision is the single most important one for interpreting the results, and its consequences are discussed in Section 6.

3. Results overview

3.1 Project types

PROJECT_TYPE	Projects	Share
QDA_PROJECT	0	0.0%
QD_PROJECT	355	91.7%
OTHER_PROJECT	31	8.0%
NOT_A_PROJECT	1	0.3%
Total	387	100.0%

Table 2: PROJECT_TYPE distribution over all repositories.

No project in the archive is a QDA_PROJECT. Not one of the 11,979 files carries a QDA file extension, so by the Step 1 rule the type cannot be assigned. This is discussed in Section 6.

3.2 Distributions to report

The task description crosses repository with project type; each populated cell is one distribution to report. Only one such cell exists in this archive:

Repository	QDA_PROJECT	QD_PROJECT
1. Harvard Murray Research Archive	none	Distribution 1 355 projects, 20 classes
2. Australian Data Archive (ADA)	none	none

Table 3: Distributions to report, by repository and project type.

4. Repository: Harvard Murray Research Archive

URL: <https://www.murray.harvard.edu>

Identifier: harvard-murray-archive (repository_id 1)

4.1 Project types found

PROJECT_TYPE	Projects	Share
QDA_PROJECT	0	0.0%
QD_PROJECT	355	92.0%
OTHER_PROJECT	31	8.0%
NOT_A_PROJECT	0	0.0%

Table 4: PROJECT_TYPE distribution for Harvard Murray Research Archive.

352 of 355 in-scope projects (QDA_PROJECT and QD_PROJECT) received a class. 2,258 of 11,978 files could be downloaded, and 1,631 of those yielded machine-readable text for the classifier to read.

4.2 Distribution of primary classes

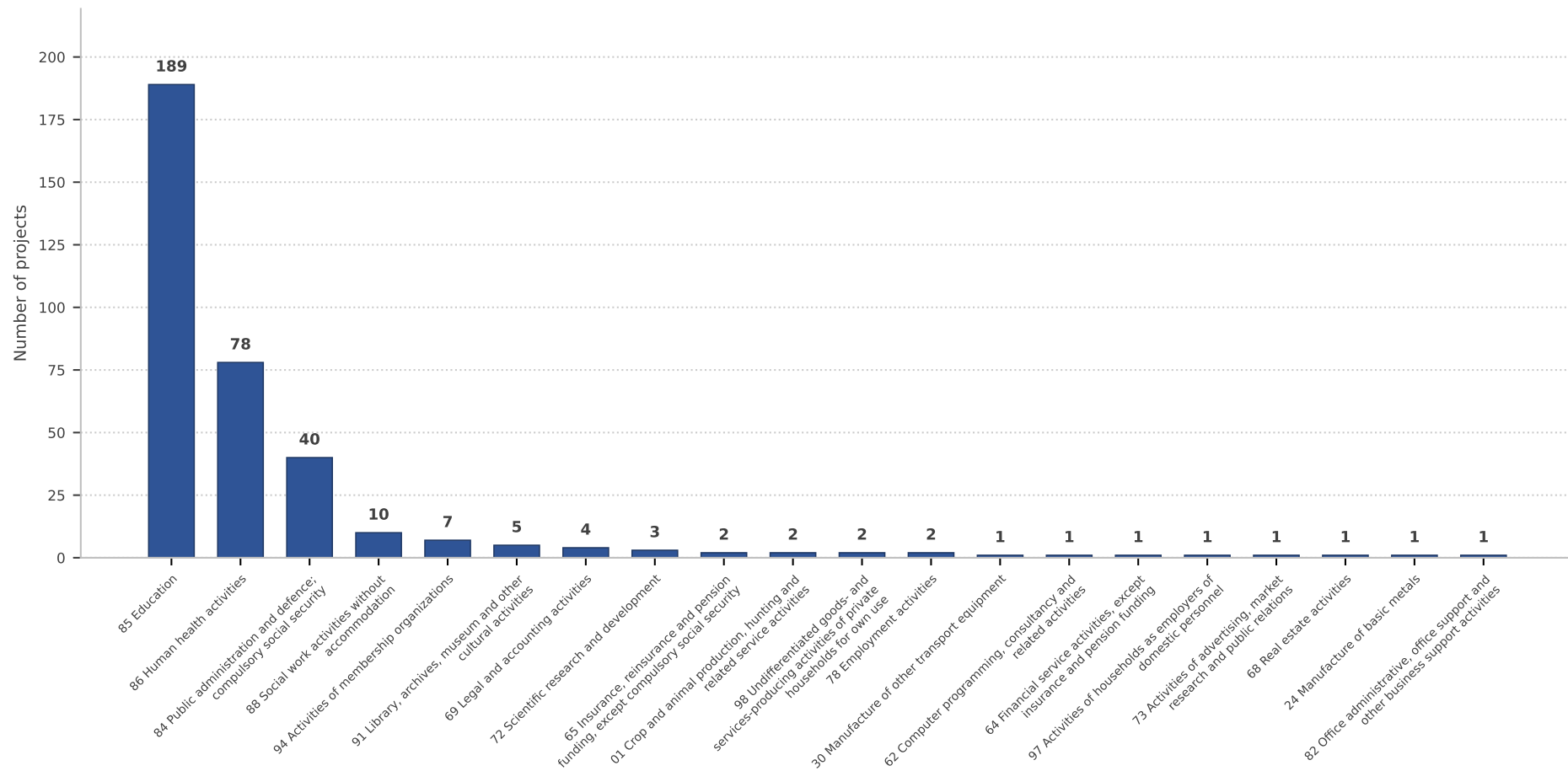


Figure 1: Histogram of the primary ISIC Rev. 5 classes identified in the Harvard Murray Research Archive, over all 352 classified projects. Bin names are the full class names; the count is printed above each bar.

4.3 Classes ranked by frequency

Rank	Section	Division	Full class name	Count	Share
1	Q	85	Education	189	53.7%
2	R	86	Human health activities	78	22.2%
3	P	84	Public administration and defence; compulsory social security	40	11.4%
4	R	88	Social work activities without accommodation	10	2.8%
5	T	94	Activities of membership organizations	7	2.0%
6	S	91	Library, archives, museum and other cultural activities	5	1.4%
7	N	69	Legal and accounting activities	4	1.1%
8	N	72	Scientific research and development	3	0.9%
9	L	65	Insurance, reinsurance and pension funding, except compulsory social security	2	0.6%
10	A	01	Crop and animal production, hunting and related service activities	2	0.6%
11	U	98	Undifferentiated goods- and services-producing activities of private households for own use	2	0.6%
12	O	78	Employment activities	2	0.6%
13	C	30	Manufacture of other transport equipment	1	0.3%
14	K	62	Computer programming, consultancy and related activities	1	0.3%
15	L	64	Financial service activities, except insurance and pension funding	1	0.3%
16	U	97	Activities of households as employers of domestic personnel	1	0.3%
17	N	73	Activities of advertising, market research and public relations	1	0.3%
18	M	68	Real estate activities	1	0.3%
19	C	24	Manufacture of basic metals	1	0.3%
20	O	82	Office administrative, office support and other business support activities	1	0.3%

Table 5: The 20 most common primary classes in Harvard Murray Research Archive, ranked, out of 20 distinct classes over 352 classified projects. The top 20 are shown.

4.4 Comments on the findings

- No file in this repository carries a QDA file extension (.qdp, .nvp, .atproj, .mx*, and so on), so by the Step 1 rule no project can be a QDA_PROJECT. The archive largely predates the REFI-QDA standard and stores its analysis material as PDF codebooks and SPSS/SAS data rather than as QDA software projects. This is a property of the data and not a gap in the pipeline: the classifier does look for those extensions.
- The primary data is dominated by .pdf (9,344), .txt (244), .doc (149). Because a single PDF is enough to make a project a QD_PROJECT, that rule sorts 92% of the repository into one bucket; the file-type rules discriminate far less here than the class taxonomy does.
- 9,720 of 11,978 files (81%) are restricted and could not be downloaded, so for those the classifier reads only the metadata and the file name. Classification quality is therefore not uniform across the repository: the projects whose files are open are classified on more evidence than the rest.

- The distribution is heavily concentrated: 87% of the classified projects fall into just three divisions, and only 20 of the 87 ISIC divisions occur at all. This mirrors the collection policy of the archive rather than a limitation of the taxonomy; it is a themed archive of social-science studies, so the manufacturing, mining and transport divisions are simply absent.
- That “Education” dominates is consistent with the corpus: the studies follow school and college populations over time, so schooling vocabulary is what the researchers themselves wrote in the metadata.
- ISIC classifies economic activities, and much of this archive is about family life, personal identity and the life course, which no ISIC division describes directly. Such projects land on the division of the institution through which they were studied — a study of mothers observed via their children’s schools scores as Education — so the classes should be read as “the sector this data speaks about”, not as a summary of the research question.
- 3 in-scope project(s) matched no lexicon term and were deliberately left unclassified rather than forced into a division, following the instruction to leave a field empty when it cannot be filled.

5. Repository: Australian Data Archive (ADA)

URL: <https://dataverse.ada.edu.au>

Identifier: ada (repository_id 2)

5.1 Project types found

PROJECT_TYPE	Projects	Share
QDA_PROJECT	0	0.0%
QD_PROJECT	0	0.0%
OTHER_PROJECT	0	0.0%
NOT_A_PROJECT	1	100.0%

Table 6: PROJECT_TYPE distribution for Australian Data Archive (ADA).

0 of 0 in-scope projects (QDA_PROJECT and QD_PROJECT) received a class. 0 of 1 files could be downloaded, and 0 of those yielded machine-readable text for the classifier to read.

5.2 Comments on the findings

- This repository could not be harvested in Part 1: its web application firewall rejected every automated request, so no project metadata and no files were ever acquired.
- The single record is the sentinel row the Part 1 pipeline writes to document that repository-level failure. It is not a research project, it carries an invented file name, and it is therefore typed NOT_A_PROJECT and excluded from classification rather than being allowed to contribute a spurious QD_PROJECT and a spurious class to the statistics.
- No distribution can be reported for this repository. This is a data acquisition limitation carried over from Part 1, not a classification result: the repository is browsable by a human but not machine-accessible.

6. Technical challenges with the data

The following observations concern the data itself rather than the programming, as requested for the ongoing reporting of this project.

6.1 The archive contains no QDA files at all

Not one of the 11,979 files carries a QDA file extension: no .qdp, no NVivo, ATLAS.ti or MAXQDA project. By the Step 1 rule there are therefore zero QDA_PROJECTs. The Murray archive largely predates the REFI-QDA interchange standard and stores its analysis material as PDF codebooks and SPSS/SAS data instead. This is a property of the data, not a gap in the pipeline: the classifier does look for those extensions, and the QDA extension list is configuration rather than hard-coded, so it can be extended when new formats appear.

6.2 The QD_PROJECT rule barely discriminates

A single PDF is enough to make a project a QD_PROJECT, and 9,342 of the files are PDFs. The rule therefore sorts 92% of the archive into one bucket. For this repository the file-type rules separate far less than the class taxonomy does, and the useful discrimination comes from the classification rather than from the typing.

6.3 Most of the base data is restricted

Only 2,258 of 11,979 files could be downloaded, and 1,631 yielded machine-readable text. The classifier reads the base data where it exists and falls back to the metadata elsewhere, so classification quality is not uniform: projects with open files are classified on more evidence than the rest. Public visibility of a repository does not imply machine accessibility, and this remains the dominant limitation carried over from Part 1.

6.4 Titles alone are often not classifiable

Many titles name only the study, for example “Woman’s Day Survey, 1984”, and contain no subject term whatsoever. Such a project still classifies from its description, but its individual files initially did not, because a file’s context was built from the title and keywords only. Including the description in the inherited context raised file-level coverage from 50% to 99.9%.

6.5 ISIC is an economic taxonomy applied to social-science data

Much of the archive is about family life, personal identity and the life course, which no ISIC division describes directly. Such projects land on the division of the institution through which they were studied: a study of mothers observed via their children’s schools scores as Education. The classes should be read as “the sector this data speaks about”, not as a summary of the research question. This is the single biggest interpretive caveat on the results, and it is a mismatch between the standard and the material rather than a defect of the classifier.

6.6 A sentinel row is not a project

Part 1 recorded ADA’s firewall block as a synthetic row carrying an invented file name, all-files-inaccessible.txt. Typed naively, its .txt extension would have made it a QD_PROJECT and contributed a spurious class to the statistics. It is detected explicitly and typed NOT_A_PROJECT, and the report states plainly that no distribution can be reported for ADA. Records that document a failure must not be allowed to read as data.

7. Conclusion

Part 2 typed all 387 projects acquired in Part 1, classified 352 of the 355 in-scope projects and 9,939 of the 9,946 primary data files against ISIC Rev. 5 at division level, and reported the resulting distributions by repository and by project type.

The dominant class is 85 Education (189 projects), followed by human health and public administration. The concentration reflects the collection policy of the Murray archive, a themed archive of longitudinal social-science studies following school and college populations, rather than any property of the taxonomy.

Two findings are worth carrying into the seeding effort itself. First, an archive can be full of qualitative research and still contain no QDA files: the interchange standard is younger than most of the material, so seeding QDArchive from existing repositories will mostly yield primary data rather than the analysis files that are of particular interest. Second, most of the material is visible but not downloadable, so the quantity of metadata that can be harvested substantially exceeds the quantity of base data that can be read.

The classifier is deterministic and its lexicon is data rather than code, so the results in this report can be reproduced exactly, and the classification can be extended to further repositories or refined without retraining anything.

8. Appendix: reproducing these results

The delivered database is `23025328-sq26-classification.db`, committed to the project repository and tagged `classification-results`. It carries the Part 1 tables extended with the `PROJECT_TYPE` and the classification, a `TAGS` table for searching, and the ISIC Rev. 5 taxonomy itself. The report, the spreadsheet and the form answers are regenerated from it by:

```
pip install -r requirements.txt
python -m part2 # all steps, end to end
python -m unittest discover -s part2/tests -t . # the test suite
```

The ISIC Rev. 5 structure was taken from the UN Statistical Commission background document at https://unstats.un.org/UNSDWebsite/statcom/session_54/documents/BG-3j-ISIC-Rev5-E.pdf, parsed to 22 sections and 87 divisions, and is stored in the repository as `data`.